DATA 266 - HW7

GAN concepts and FashionMNIST AutoEncoders

Siva Surya Chandran

019130215

Step 0: Personal parameters

SID4	SEED	SLICE	HP_ID	CLS_A	CLS_B
3215	3215	215	5	5	2

I use the same personal-parameter convention as the earlier homeworks. HW7 does not define a separate HP_ID experiment. The two required models, FC-AE and VAE, are the two models trained for this assignment.

Models: Fully Connected AutoEncoder (FC-AE) and Variational AutoEncoder (VAE)

This notebook is meant to be run from the first cell to the last. I trained the two models required for the assignment, so I did not add a separate hyperparameter experiment.

1. GAN concepts

1.A Fundamental architecture

A Generative Adversarial Network, or GAN, has two neural networks that compete with each other. The generator G(z) starts with a random latent vector and turns it into a fake sample. The discriminator D(x) looks at a sample and predicts whether it is real or generated. During training, the discriminator learns from real and fake examples, while the generator uses the discriminator's feedback to make better fake examples.

The original minimax objective is

$$\min_G \max_D V(D,G) = \mathbb{E}_{x \sim p_{data}}[\log D(x)] + \mathbb{E}_{z \sim p(z)}[\log(1 - D(G(z)))]$$

In theory, training reaches an equilibrium when the generator's distribution matches the real data distribution, p_g=p_{data}. At that point, the discriminator cannot tell the two distributions apart and outputs 1/2 for either one. In practice, the non-saturating generator loss -\mathbb E_z[\log D(G(z))] is often used because it gives the generator stronger gradients at the beginning of training.

1.B Mode collapse

Mode collapse happens when the generator keeps producing only a few kinds of samples, even when it receives different random inputs. The images may look realistic, but they do not represent the full variety of the training data. This happens because the generator can find one small group of outputs that fools the discriminator and then keep using that shortcut. The two networks are also being optimized at the same time, so the changing discriminator can create unstable gradients and make this behavior worse.

Two mitigation strategies are:

1. **Minibatch discrimination.** The discriminator is allowed to compare examples within the same minibatch instead of judging every image completely independently. If the generator creates six nearly identical images, the discriminator can notice that the batch lacks variety and penalize it. This gives the generator a reason to produce different kinds of outputs.
2. **Wasserstein GAN with gradient penalty (WGAN-GP).** WGAN changes the discriminator into a critic that estimates the Wasserstein distance between the real and generated distributions. This distance usually gives more useful gradients when the two distributions are still far apart. WGAN-GP also adds a gradient penalty on interpolated real and fake samples, which is more stable than simply clipping the critic's weights.

Other approaches include feature matching, adding noise to the discriminator, unrolled optimization, and mode-seeking losses that directly reward diversity.

1.C Three variants

StyleGAN is another major extension. It sends different parts of a transformed latent code into different generator layers, which gives better control over high-level features such as pose and lower-level details such as texture.

Interpretation and limitations

The FC-AE only tries to reproduce each input image, so it gives the lower reconstruction error in this experiment. Its outputs can still look blurry when the original clothing item has soft or ambiguous edges. The VAE adds a KL term that keeps its latent codes close to a standard normal distribution. That makes it possible to generate new images from random latent vectors, but the extra regularization makes reconstruction less sharp. I use test MSE and PSNR to compare reconstruction quality, and I also check the error for each clothing class.

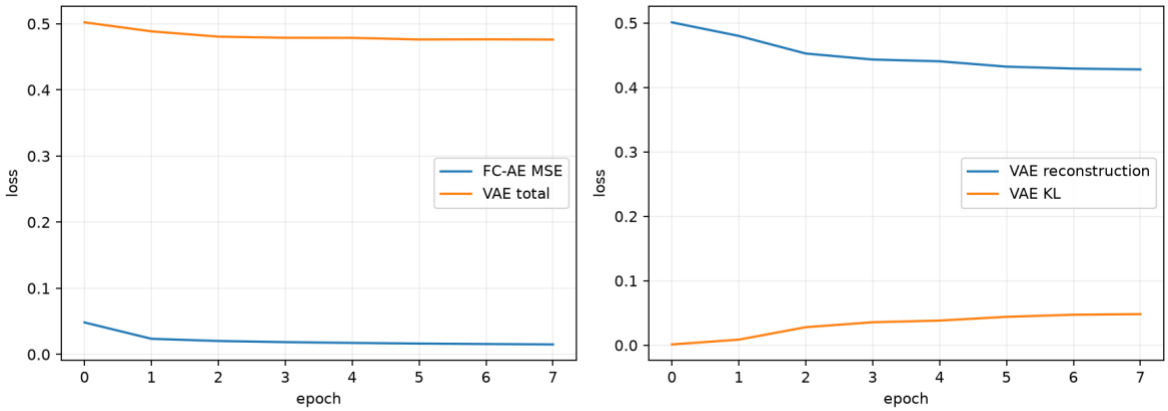
The worst examples are not always meaningless images. Pixel MSE is sensitive to small shifts and texture differences, and FashionMNIST is only 28 by 28 pixels. A better follow-up evaluation could include SSIM or a classifier-based perceptual score. I did not add a latent-size or KL-weight sweep because the assignment asks for the two model types rather than another hyperparameter comparison.

Experimental results

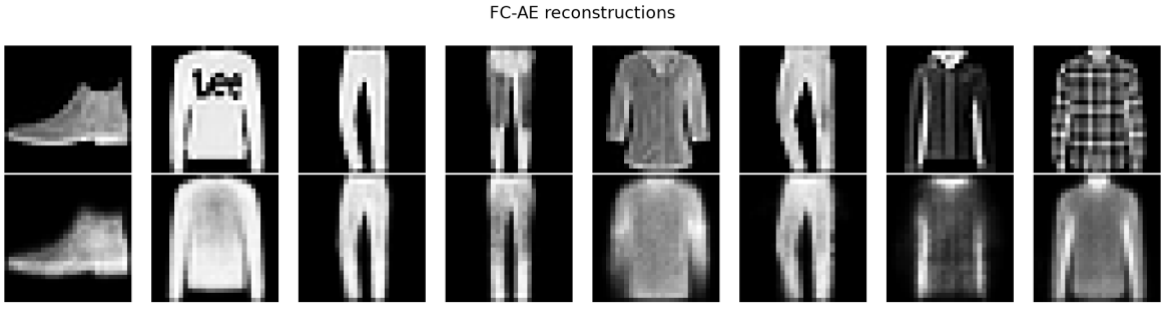
I used the standard FashionMNIST split with 60,000 training images and 10,000 test images. Both models used batch size 256, Adam with learning rate 0.001, 8 epochs, and SEED 3215 on Apple MPS.

Model	Latent size	Parameters	Final train loss	Test MSE	Test PSNR
FC-AE	32	476,720	0.014989	0.014938	18.257 dB
VAE	16	474,672	0.476111	0.062484	12.042 dB

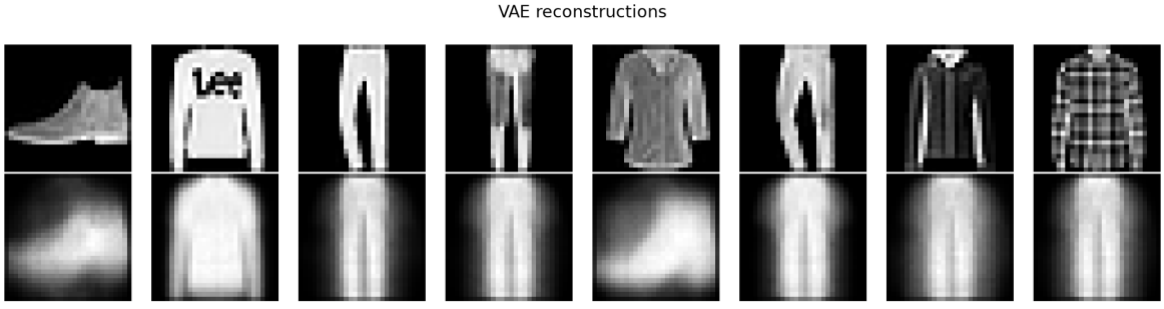
The FC-AE had lower reconstruction error because it focuses only on reproducing the input pixels. The VAE gives up some reconstruction sharpness in exchange for a smoother latent space that can be sampled to create new images.



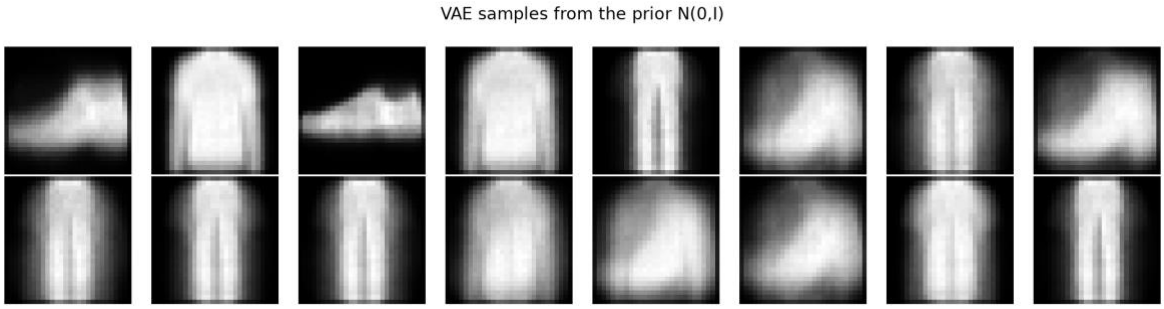
Training losses.



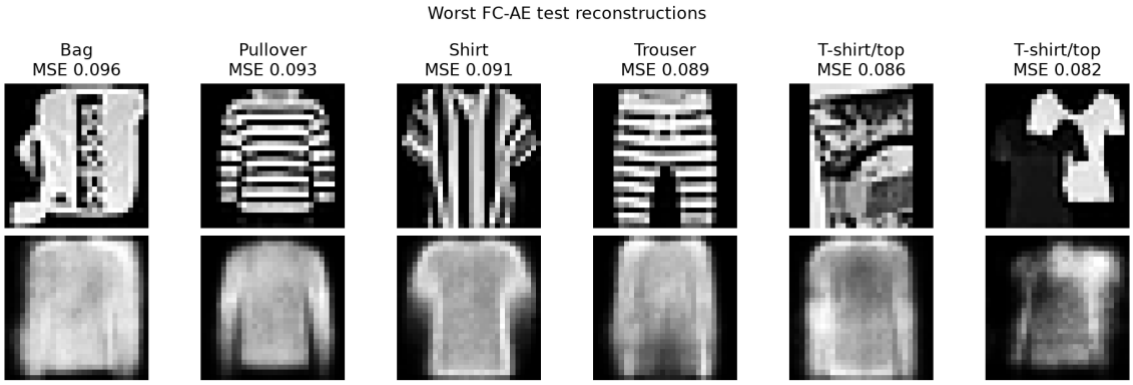
FC-AE reconstructions.



VAE reconstructions.



VAE samples generated from random latent vectors.



Six high-error FC-AE reconstructions.

AI-use appendix

I used an assistant to help organize the notebook, draft the GAN explanations, write some of the PyTorch training code, and check the required files. I wrote the final interpretation, chose the two required models, checked the formulas, and inspected the results and figures myself.

One issue in the first version was the way the VAE loss terms were scaled. The first run had a KL term of only 0.000007 and a test PSNR of 10.616 dB, so the model was barely using the variational regularizer. I found this by looking at the separate reconstruction and KL values printed during training. I changed the reconstruction term to mean binary cross-entropy while keeping the mean KL term. After rerunning, the final KL loss was 0.048201, the reconstruction loss was 0.427910, and the test PSNR was 12.042 dB.